

Lab 10

LabSection.java

```
LabSection.java | RegisterRunnable.j... | UnregisterRunnabl... | progDemo.java |
1- public class LabSection {
2-
3-     private int numOfReg;
4-
5-     public LabSection(int r) {
6-         numOfReg = r;
7-     }
8-
9-     public synchronized void register() {
10-
11-         try {
12-             while (numOfReg >= 25)
13-                 wait();
14-         } catch (Exception e) {
15-         }
16-
17-         int updNoOfReg = numOfReg + 1;
18-
19-         System.out.println("Register-request, registered students: "
20-             + updNoOfReg);
21-
22-         numOfReg = updNoOfReg;
23-     }
24-
25-     public synchronized void unregister() {
26-
27-         int updNoOfReg = numOfReg - 1;
28-
29-         System.out.println("Unregister-request, registered students: "
30-             + updNoOfReg);
31-
32-         numOfReg = updNoOfReg;
33-
34-         notifyAll();
35-     }
36-
37-     public double numOfRegisteredStudents() {
38-         return numOfReg;
39-     }
40- }
41
```

RegisterRunnable.java

```
LabSection.java | RegisterRunnable.j... | UnregisterRunnabl... | progDemo.java |
1- public class RegisterRunnable implements Runnable {
2-
3-     private static final int DELAY = 200;
4-     private LabSection osLabGroup;
5-
6-     public RegisterRunnable(LabSection osLab) {
7-         osLabGroup = osLab;
8-     }
9-
10-    public void run() {
11-
12-        try {
13-            osLabGroup.register();
14-            Thread.sleep(DELAY);
15-        } catch (InterruptedException exception) {
16-        }
17-    }
18- }
```

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UnregisterRunnable.java

```
LabSection.java : RegisterRunnable.j... : UnregisterRunnabl... : progDemo.java :
1 public class UnregisterRunnable implements Runnable {
2
3     private static final int DELAY = 200;
4     private LabSection osLabGroup;
5
6     public UnregisterRunnable(LabSection osLab) {
7         osLabGroup = osLab;
8     }
9
10    public void run() {
11
12        try {
13            osLabGroup.unregister();
14            Thread.sleep(DELAY);
15        } catch (InterruptedException exception) {
16        }
17    }
18 }
```

progDemo.java

```
LabSection.java : RegisterRunnable.j... : UnregisterRunnabl... : progDemo.java :
1 public class progDemo {
2
3     public static void main(String[] args) {
4
5         LabSection osLab = new LabSection(25);
6
7         for (int i = 1; i <= 25; i++) {
8
9             RegisterRunnable reRun = new RegisterRunnable(osLab);
10            UnregisterRunnable unRegRun = new UnregisterRunnable(osLab);
11
12            Thread regThrd = new Thread(reRun);
13            Thread unRegThrd = new Thread(unRegRun);
14
15            regThrd.start();
16            unRegThrd.start();
17        }
18    }
19 }
```

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The OutPut:

[illegible]